

MIDTERM EXAMINATION 2026

1. What is the time complexity of QuickSort in the average case?

- A. $O(n)$
- B. $O(n \log n)$
- C. $O(n^2)$
- D. $O(1)$

2. Which data structure operates on a First-In, First-Out (FIFO) basis?

- A. Stack
- B. Tree
- C. Queue
- D. Graph

3. What is the derivative of $f(x) = x^2$ with respect to x ?

- A. $2x$
- B. x
- C. 2
- D. x^3

4. What is the approximate speed of light in a vacuum?

- A. 3×10^8 m/s
- B. 1×10^6 m/s
- C. 300 m/s
- D. Infinite